SRS Report

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	20
Problem understanding & Analysis	7					
Implementation	8					
Report writing	5					
					Total obtained mark	
Comments						

Semester: Fall 2026

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Software Requirements Specification (SRS)

CareGuide Rural AI: An AI-Powered Rural Healthcare Assistance System

1. Introduction

CareGuide Rural AI is a web-based healthcare information and navigation assistant designed to help rural users in Bangladesh access healthcare information more easily. The system will provide a Bangla-first interface through which users can ask healthcare-related questions, search for hospitals and healthcare services, and receive simple explanations of supported medical documents. It will use artificial intelligence (AI) to understand user queries, retrieve relevant information from a curated healthcare knowledge base, process medical document images, and support voice-based interaction.

1.1 Problem Statement

Rural users may face difficulties accessing healthcare information because relevant information is often scattered across different sources, presented in technical language, or difficult to find through conventional search methods. Finding suitable hospitals, departments, and diagnostic services may require users to search multiple websites or directories. Additionally, users may struggle to understand medical reports containing unfamiliar medical terms, particularly when explanations in simple Bangla are unavailable. Users who are less comfortable typing may also face difficulties using conventional web-based platforms.

Traditional healthcare directories mainly provide predefined information and require users to search for and interpret the results themselves. General-purpose AI systems may also provide information that is not sufficiently relevant to local healthcare facilities or the project's selected knowledge sources.

Therefore, a unified web-based solution is needed to make healthcare information easier to access and understand. AI is required to interpret natural-language queries, retrieve relevant information from a curated healthcare knowledge base, explain readable medical document content in simple Bangla, and support voice interaction. CareGuide Rural AI aims to address these challenges by combining healthcare question answering, hospital and service search, and medical document explanation in a single platform.

1.2 Purpose

The purpose of CareGuide Rural AI is to develop an AI-powered web application that improves access to general healthcare information and healthcare service navigation for rural users. The primary objectives of the software are:

1. To provide AI-powered question answering for healthcare-related queries in Bangla.
2. To retrieve relevant information from a curated healthcare knowledge base and generate clear, informative responses.
3. To help users find hospitals, departments, and diagnostic services based on their location and requirements.
4. To support Bangla voice interaction through speech-to-text and text-to-speech technologies.
5. To process supported medical document images and explain their readable contents in simple Bangla.
6. To provide relevant information sources where available, helping users verify the information presented.
7. To evaluate the accuracy, relevance, and usability of the AI-generated responses and system features.

1.3 Scope

The scope of CareGuide Rural AI includes the development of a web application that integrates healthcare information retrieval, AI-assisted question answering, hospital and service search, medical document explanation, and Bangla voice interaction. The initial implementation will focus on a selected geographical area and a curated healthcare knowledge base to keep the project manageable and ensure that the available information can be maintained.

In-Scope Features

- **Healthcare Question Answering:** Users can submit healthcare-related questions in Bangla and receive simple, informative responses.
- **Knowledge-Based Information Retrieval:** The system will retrieve relevant content from a curated healthcare knowledge base using a Retrieval-Augmented Generation (RAG) approach.
- **Hospital and Service Search:** Users can search for hospitals, departments, and diagnostic services based on available location and service information.
- **Bangla Voice Interaction:** Users can submit spoken queries and listen to generated responses using speech-to-text and text-to-speech technologies.
- **Medical Document Explanation:** Users can upload supported medical document images and receive simple Bangla explanations of readable content.
- **AI Agent and Tool Selection:** The AI workflow will select appropriate predefined tools, such as healthcare information retrieval and hospital search, based on the user's query.
- **Source Display:** The system will display supporting sources for healthcare answers where available.
- **Information Management:** Authorized administrators can manage hospital records and update the healthcare information maintained by the system.

Out-of-Scope Features

- Automated disease diagnosis or clinical decision-making.
- Prescription generation, medication recommendations, or dosage modification.

- Emergency diagnosis or replacement of emergency medical services.
- Guaranteed interpretation of handwritten prescriptions or unclear medical documents.
- Direct integration with hospital electronic health record systems.
- Online appointment booking and payment processing in the initial version.
- Nationwide healthcare facility coverage in the initial implementation.
- Training a large language model from scratch.

AI Limitations and Safety Boundaries

The AI components will support healthcare information access, information retrieval, document understanding, and natural-language interaction. The system will not diagnose diseases, prescribe medication, recommend changes to treatment, or replace qualified healthcare professionals. AI-generated information may contain errors; therefore, users will be advised to verify important medical information with qualified professionals. If the available information is insufficient or a document cannot be read reliably, the system should communicate this limitation rather than present an uncertain interpretation as fact.

2. Design and Implementation Constraints

CareGuide Rural AI will be developed using a web-based client-server architecture with an AI-powered backend. The system will integrate a relational database, vector search, pretrained AI services, speech processing, and interactive map technologies. Implementation will depend on internet connectivity, external API availability, usage limits, and the quality of available healthcare data.

2.1 Programming Language

The project will use JavaScript for frontend development and Python for backend and AI-related functionalities.

- **JavaScript:** Used with React.js to develop the interactive web interface for healthcare queries, hospital searches, voice interaction, and medical document uploads.
- **React.js:** Used to build reusable UI components and manage user interactions.
- **Tailwind CSS:** Used to develop a responsive interface for desktop and mobile devices.
- **Python:** Used for backend logic and AI integration because of its extensive support for AI and natural-language processing libraries.
- **FastAPI:** Used to develop RESTful APIs connecting the frontend with the database and AI services.
- **LangChain and LangGraph**: LangChain will support RAG and LLM integration, while LangGraph will manage the AI agent workflow and controlled tool execution.

Justification: This technology stack separates the user interface from backend and AI processing. JavaScript and React.js support interactive web development, while Python provides suitable

libraries for implementing healthcare information retrieval, AI-based question answering, and document processing.

2.2 Database & Servers

The system will use PostgreSQL as its primary relational database and pgvector for vector storage and similarity search. FastAPI will manage communication between the frontend, database, and AI services.

- **PostgreSQL:** Used to store hospital details, locations, departments, diagnostic services, and relevant application records.
- **pgvector:** Used to store embeddings of healthcare knowledge documents and perform similarity searches for the RAG pipeline.
- **FastAPI Server:** Provides RESTful endpoints for healthcare queries, hospital and service searches, document processing, and AI service integration.
- **Frontend–Backend Communication:** The React.js frontend will communicate with FastAPI through HTTP requests and JSON-based REST APIs. Uploaded document images will be sent through file-upload requests.
- **Data Retrieval:** SQL queries will retrieve structured hospital and service information, while vector similarity search will retrieve relevant healthcare knowledge passages.
- **Map Services:** Leaflet will provide the interactive map interface, and OpenStreetMap will supply map data for displaying hospital locations and helping users explore healthcare facilities without requiring a paid map API.
- **Configuration and Security:** API keys and sensitive configuration values will be stored in environment variables rather than hardcoded into the source code.

Justification: PostgreSQL and pgvector allow structured records and vector embeddings to be managed within one database system. FastAPI connects the application components, while Leaflet and OpenStreetMap support map-based hospital navigation with reduced infrastructure costs.

2.3 AI Models & Services

- **Gemini 3.8 Flash:** Primary candidate LLM for Bangla healthcare responses and query understanding.
- **Gemini 3.5 Flash Lite:** Fallback LLM, tested for response generation when the primary model is unavailable.
- **Gemini Function Calling:** Selects predefined tools such as `rag_search`, `hospital_search`, and `report_explanation` based on user queries.
- **LangGraph:** Manages the AI agent workflow, including tool selection, execution, and response generation. Basic integration has been tested using mock data.
- **RAG:** Planned for retrieving relevant information from a trusted healthcare knowledge base.

- **BGE-M3:** Planned for generating multilingual embeddings for semantic search.
- **LangChain:** Planned for integrating the LLM and retrieval components in the RAG pipeline.
- **PostgreSQL with pgvector:** Planned for storing healthcare data and vector embeddings.
- **STT — Gemini 3.5 Transcribe:** Tested for converting spoken Bangla queries into text.
- **TTS — Gemini 3.8 Flash TTS:** Tested for converting generated Bangla responses into audio.
- **Medical Report Processing:** EasyOCR and Gemini are planned for extracting and explaining medical report text.
- **Cost Constraint:** Available API quotas will be used for prototyping and testing, subject to usage limits and pricing.

- **Safety Constraints:** The AI system will provide informational guidance only and will not diagnose diseases, prescribe medicines, or recommend changes to medication dosages.

Justification: These technologies provide multilingual question answering, context-based information retrieval, document explanation, and voice interaction without requiring the project to train a large language model from scratch.

3. User Classes and Characteristics

- **General Users:** Rural residents or caregivers who use the system to obtain healthcare information in Bangla, find nearby hospitals and diagnostic services, and understand medical report content.
- **System Administrator:** Manages hospital and diagnostic service records, maintains the healthcare knowledge base, and oversees system configuration.
- **Healthcare Information Reviewer:** Reviews healthcare information and reference materials to ensure their accuracy and relevance. In the initial version, this role may be performed by the system administrator.

4. Requirement Elicitation

Requirements for CareGuide Rural AI will be identified through interviews and questionnaires involving potential users and, where possible, healthcare professionals. The collected feedback will help determine user needs and refine the system's features, usability, accessibility, and safety requirements.

4.1 Elicitation Techniques

The requirements for CareGuide Rural AI were collected using the following techniques:

- **Interviews:** Interview questions were prepared to understand users' difficulties in accessing healthcare information, finding hospitals, and understanding medical reports.
- **Questionnaires:** Questions were designed to identify users' preferences regarding Bangla responses, voice interaction, hospital search, and medical report explanations.
- **Requirement Analysis:** The identified user needs were analyzed to determine the proposed system's functional and non-functional requirements.
- **Participants:** Potential users, including rural residents and caregivers, were considered the primary target group. Healthcare professionals were also considered potential participants for gathering additional feedback.
- **Schedule:** Requirement elicitation activities were conducted during the project requirements analysis phase. The specific dates and participant details should be included based on the actual activities performed.

4.2 Elicitation Evidence

Evidence of requirement elicitation was collected through interviews with potential users of CareGuide Rural AI. The evidence includes the interview questions and summarized interview notes documenting users' healthcare information needs, difficulties in finding hospitals, and preferences for the proposed system. The interview records are included in the appendix to support the identified requirements.

Appendix A: Interview Questions

1. Where do you usually obtain healthcare information?

2. Have you ever faced difficulties finding hospitals or diagnostic centres? If yes, what difficulties did you experience?

6. Would receiving healthcare information in simple Bangla be helpful to you? Why?

3. In an emergency, would you like the system to provide quick access to nearby healthcare facilities and emergency contact information?

4. Would it be helpful if the system showed directions and approximate distances to your selected hospital?

5. Would you like to access your previous healthcare questions and hospital searches?

6. Would you find it useful to view previous medical report explanations later, provided they are stored securely with your consent?

7. Would you like the option to report incorrect healthcare information or outdated hospital and diagnostic service details?

8. What additional features would you like to see in the proposed system?

Appendix B: Interview Notes and Response Summary

Q1. Where do you usually obtain healthcare information?

Answer: I usually get healthcare information from doctors, family members, and online searches.

Q2. Have you ever faced difficulties finding hospitals or diagnostic centres? If yes, what difficulties did you experience?

Answer: Yes, sometimes it is difficult to find the correct hospital location, contact number, and available services.

Q3. Would receiving healthcare information in simple Bangla be helpful to you? Why?

Answer: Yes, simple Bangla would help me understand healthcare information more easily.

Q4. In an emergency, would you like the system to provide quick access to nearby healthcare facilities and emergency contact information?

Answer: Yes, it would help me quickly find nearby hospitals and emergency contact numbers.

Q5. Would it be helpful if the system showed directions and approximate distances to your selected hospital?

Answer: Yes, directions and distance would help me choose a suitable hospital and plan my journey.

Q6. Would you like to access your previous healthcare questions and hospital searches?

Answer: Yes, I would like to check my previous searches without searching again.

Q7. Would you find it useful to view previous medical report explanations later, provided they are stored securely with your consent?

Answer: Yes, it would be useful to review previous explanations when needed, provided my information remains private.

Q8. Would you like the option to report incorrect healthcare information or outdated hospital and diagnostic service details?

Answer: Yes, I would like to report incorrect information so it can be reviewed and updated.

Q9. What additional features would you like to see in the proposed system?

Answer: I would like updated hospital information, and a simple interface that works well with slow internet.

Appendix C: Interview Details and Evidence Collection

Date of Evidence Collection: 11 October 2026

On 11 October 2026, we visited a rural community to collect requirements and evidence for our proposed project, CareGuide Rural AI. We communicated directly with local people to understand their healthcare-related problems, difficulties in finding nearby hospitals, challenges in understanding medical reports, and the need for Bangla-based healthcare information.

We collected information through discussions, interviews, and observations. We also documented the collected evidence to identify user needs and determine the essential features of our proposed system.

Evidence Collected:

- Photographs of the field visit.
- Interview questions and responses.
- Notes on users' healthcare-related problems.
- Observations and suggestions from local people.







4.3 Key Findings

The interview responses highlighted the following potential user needs for CareGuide Rural AI:

- **Accessible Healthcare Information:** Users need reliable healthcare information presented in simple Bangla for easier understanding.
- **Hospital Search:** Users need easy access to hospital locations, contact numbers, and available services.
- **Emergency Support:** Quick access to nearby healthcare facilities and emergency contact information is important.
- **Search and Report History:** Users may benefit from revisiting previous searches and medical report explanations.
- **Information Accuracy:** Users need a way to report incorrect healthcare information and outdated facility details.
- **Usability and Accessibility:** Voice input, a simple interface, and support for slow internet connections may improve usability.
- **Privacy:** Medical report history should be stored only with user consent and appropriate privacy protection.

These findings helped identify potential functional and non-functional requirements for the proposed system. The final requirements should be prioritized according to the actual interview responses and project feasibility.

4.4 Requirement Changes After Elicitation

The interview findings were reviewed to identify changes to the requirements of CareGuide Rural AI. The following table presents three important proposed changes based on user needs.

ID	Change	Before Elicitation	After Elicitation	Source / Reason
FR-01	New	No feature for reporting incorrect or outdated hospital information.	The system shall allow users to report incorrect healthcare information and outdated hospital details for review.	Users need a way to report inaccurate information.
FR-02	New	No dedicated emergency facility search feature.	The system shall help users find nearby healthcare facilities and available emergency contact information.	Users need quick access to emergency healthcare information.
FR-03	New	Previous searches and report explanations are not saved.	The system may provide recent search history and optional report history with user consent.	Users may need to revisit previous searches and report explanations.

5. Functional Requirements (FR)

5.1 User Access and Interaction

ID	Requirement Name	Description	Users	Priority	Assignee
FR-01	User Interface	The system shall provide a simple, accessible, mobile-friendly interface suitable for rural users.	General User	High	Frontend Team
FR-02	Bangla Language Support	The system shall allow users to interact with the application in Bangla.	General User	High	Frontend Team
FR-03	Healthcare Question Input	The system shall allow users to enter healthcare-related questions through a text input field.	General User	High	Frontend Team
FR-04	Conversation Display	The system shall display user queries and AI-generated responses in a readable conversation interface.	General User	High	Frontend Team
FR-05	Conversation Context	The system shall use relevant conversation context to support follow-up questions within the same conversation.	General User	Medium	AI/Backend Team

5.2 AI Healthcare Question Answering

ID	Requirement Name	Description	Users	Priority	Assignee
FR-06	Natural Language Understanding	The system shall interpret healthcare-related questions expressed in natural Bangla.	General User	High	AI Team
FR-07	Healthcare Information Generation	The system shall generate clear, easy-to-understand responses to general healthcare questions.	General User	High	AI Team
FR-08	Trusted Knowledge Retrieval	The system shall retrieve relevant passages from a curated healthcare knowledge base to support answers.	General User	High	AI Team

FR-09	Source References	The system shall display available source references for healthcare answers generated using retrieved documents.	General User	High	AI Team
FR-10	Semantic Search	The system shall retrieve relevant healthcare information based on semantic similarity rather than relying only on exact keyword matching.	General User	High	AI/Backend Team
FR-11	Unavailable Information Handling	The system shall inform users when sufficient verified information is unavailable rather than inventing unsupported answers.	General User	High	AI Team
FR-12	Follow-up Question Support	The system shall allow users to ask related follow-up questions about previously discussed healthcare information.	General User	Medium	AI Team

5.3 AI Agent and Tool Management

ID	Requirement Name	Description	Users	Priority	Assignee
FR-13	Query Classification	The system shall identify the appropriate workflow for healthcare questions, facility searches, and medical report explanations.	General User	High	AI Team
FR-14	Function Calling	The system shall use LLM function calling to select predefined tools and generate the required tool arguments.	General User	High	AI Team
FR-15	RAG Tool Execution	The system shall execute the healthcare retrieval tool when a query requires information from the healthcare knowledge base.	General User	High	AI/Backend Team
FR-16	Hospital Search Tool Execution	The system shall execute the facility search tool when a user requests hospitals or healthcare services.	General User	High	Backend Team

FR-17	Medical Report Tool Execution	The system shall execute the report-processing workflow when a user requests an explanation of an uploaded medical report.	General User	High	AI/Backend Team
FR-18	Tool Error Handling	The system shall handle tool failures or unavailable services and provide an appropriate response without presenting fabricated results.	General User	High	AI/Backend Team
FR-19	LLM Fallback Handling	The system shall attempt to use a configured fallback LLM when the primary model is unavailable, subject to retry policies and API availability.	General User	Medium	AI/Backend Team

5.4 Voice Interaction

ID	Requirement Name	Description	Users	Priority	Assignee
FR-20	Bangla Speech Input	The system shall allow users to submit spoken Bangla queries through a voice-recording interface.	General User	High	Frontend Team
FR-21	Speech-to-Text Conversion	The system shall convert recorded speech into text using a suitable speech-to-text service.	General User	High	AI/Backend Team
FR-22	Text-to-Speech Conversion	The system shall convert AI-generated responses into audio using a suitable text-to-speech service.	General User	High	AI/Backend Team
FR-23	Voice Processing Error Handling	The system shall inform users when speech recognition or audio generation fails and provide an appropriate alternative, such as text interaction.	General User	Medium	AI/Backend Team

5.5 Hospital and Healthcare Service Search

ID	Requirement Name	Description	Users	Priority	Assignee
FR-24	Location-Based Search	The system shall allow users to search for healthcare facilities by entering a location or area name.	General User	High	Backend Team
FR-25	Service-Based Search	The system shall allow users to search for facilities by healthcare service or specialty, such as child healthcare.	General User	High	Backend Team
FR-26	Facility Information Display	The system shall display available facility details, including name, address, contact information, and services.	General User	High	Backend Team
FR-27	No-Result Handling	The system shall inform users when no matching facility is found in the available database.	General User	High	Backend Team
FR-28	Emergency Contact Information	The system shall display available, verified emergency contact information and relevant facilities when such information is stored in the system.	General User	High	Backend Team

5.6 Medical Report Explanation

ID	Requirement Name	Description	Users	Priority	Assignee
FR-29	Report Upload	The system shall allow users to upload supported medical report files or images.	General User	High	Frontend/Backend Team
FR-30	File Validation	The system shall validate uploaded file types, sizes, and basic file integrity before processing.	General User	High	Backend Team
FR-31	Report Text Extraction	The system shall extract readable text from supported report images using OCR or	General User	High	AI/Backend Team

		an appropriate vision service.			
FR-32	Report Explanation	The system shall explain extracted medical report text in simple Bangla.	General User	High	AI Team
FR-33	Unreadable Report Handling	The system shall inform users when a report is unclear, incomplete, or cannot be processed reliably.	General User	High	AI/Backend Team

5.7 User History and Feedback

ID	Requirement Name	Description	Users	Priority	Assignee
FR-34	Conversation History	The system shall allow users to revisit saved conversations if conversation-history functionality is enabled.	General User	Medium	Frontend/Backend Team
FR-35	Optional Report History	The system shall allow users to revisit saved report explanations only with appropriate user consent and access controls.	General User	Medium	Backend Team
FR-36	Incorrect Information Reporting	The system shall allow users to report incorrect healthcare information or outdated facility details for review.	General User	Medium	Frontend/Backend Team

5.8 Administration and Knowledge Base Management

ID	Requirement Name	Description	Users	Priority	Assignee
FR-37	Admin Authentication	The system shall restrict administrative functions to authenticated, authorized administrators.	Admin	High	Backend Team

FR-38	Facility Management	The system shall allow authorized administrators to add, view, update, and remove healthcare facility records.	Admin	High	Backend Team
FR-39	Healthcare Service Management	The system shall allow authorized administrators to manage facility service and specialty information.	Admin	High	Backend Team
FR-40	Contact Information Management	The system shall allow authorized administrators to update facility addresses and contact details.	Admin	High	Backend Team
FR-41	Healthcare Knowledge Base Management	The system shall allow authorized administrators to add, update, or remove approved healthcare documents used by the RAG pipeline.	Admin	High	AI/Backend Team
FR-42	User Report Review	The system shall allow authorized administrators to review and manage user-submitted reports of inaccurate or outdated information.	Admin	Medium	Backend Team

6. Non-Functional Requirements (NFR)

The following are the main non-functional requirements of CareGuide Rural AI.

ID	Requirement Name	Description	Priority	Assignee
NFR-01	Usability	The system shall provide a simple, user-friendly interface with clear Bangla text and easy navigation.	High	Frontend Team
NFR-02	Performance	The system shall process user requests and return responses within a reasonable time under normal operating conditions.	High	Backend Team
NFR-03	Security and Privacy	The system shall protect API keys, restrict unauthorized access, and handle user queries and medical reports securely.	High	Backend Team
NFR-04	Reliability	The system shall handle API failures and processing errors gracefully and display appropriate error messages.	High	Backend/AI Team

NFR-05	AI Response Quality and Safety	The system shall provide relevant, understandable healthcare information, avoid unsupported medical claims, and communicate when verified information is unavailable.	High	AI Team
NFR-06	Compatibility	The system shall support commonly used modern web browsers and responsive layouts on desktop and mobile devices.	High	Frontend Team
NFR-07	Maintainability	The system shall use modular code and documented components to simplify testing, debugging, and future updates.	Medium	Development Team
NFR-08	Cost Efficiency	The system shall manage AI API usage efficiently within the available budget and applicable API quotas.	High	AI/Backend Team

7.1 Use Case Diagram

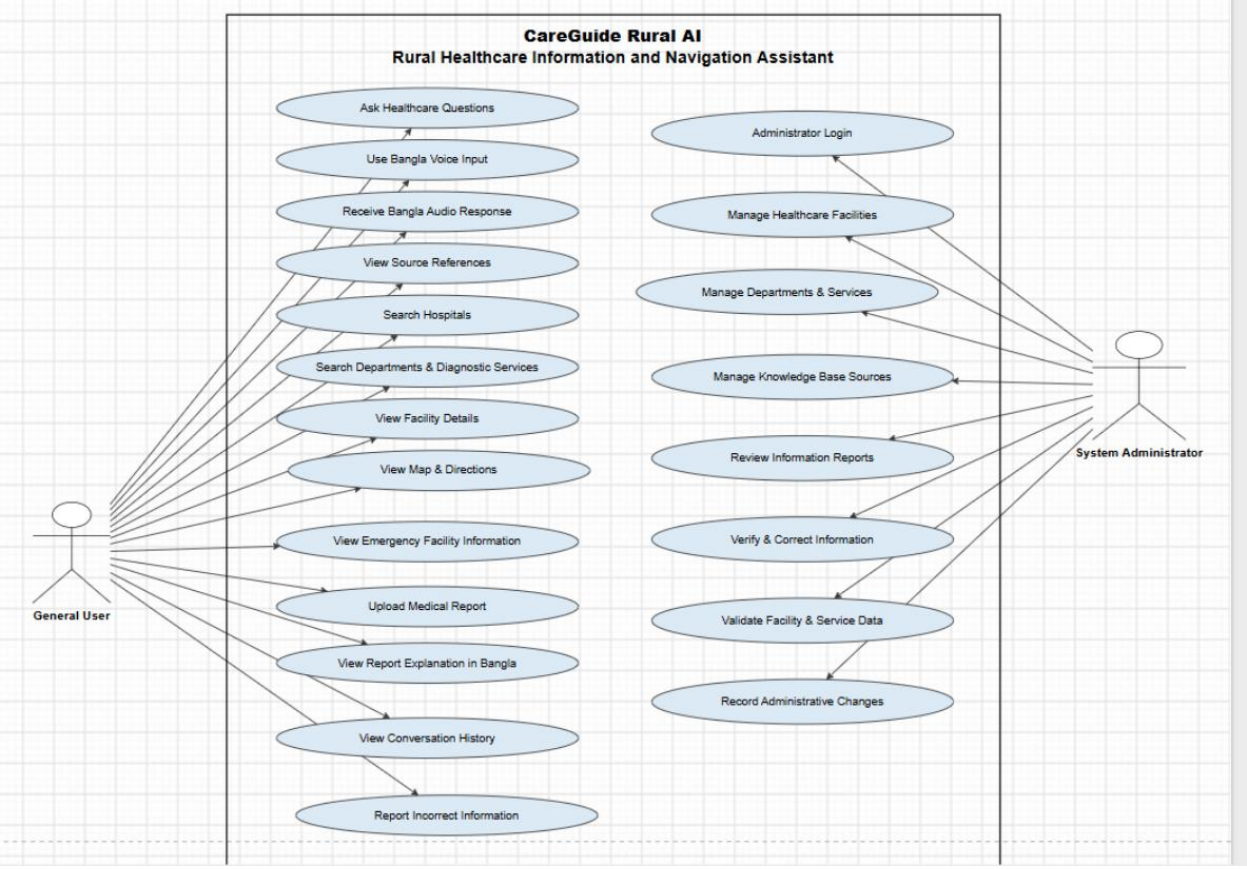
The Use Case Diagram illustrates the interactions between the two main actors of CareGuide Rural AI: **General User** and **System Administrator**. It represents the major functionalities available to each actor within the system.

General User:

The General User interacts with CareGuide Rural AI to access healthcare-related services. Users can ask healthcare questions, use Bangla voice input, receive audio responses, view source references, search hospitals and diagnostic services, view facility details and map directions, and access emergency facility information. They can also upload medical reports, view simple Bangla explanations, access conversation history, and report incorrect information.

System Administrator:

The System Administrator is responsible for managing and maintaining the CareGuide Rural AI system. The administrator can log in using authorized credentials, manage healthcare facilities, departments, and diagnostic services, and maintain knowledge base sources. Additionally, the administrator can review user-submitted information reports, verify and correct records, validate facility information, and record administrative changes.



7.2 Use Case Descriptions

Use Case Descriptions explain how users interact with the CareGuide Rural AI system. The following four important high-priority use cases describe the Main Success Flow.

7.2.1 Ask Healthcare Questions

Field	Description
Use Case Name	Ask Healthcare Questions
Primary Actor	General User
Priority	High
Description	Allows users to ask healthcare-related questions and receive answers in simple Bangla.

Main Success Flow	1. User opens the system. 2. User enters a healthcare-related question. 3. System processes the question. 4. RAG retrieves relevant information from the knowledge base. 5. LLM generates an answer. 6. System displays the answer with source references.

7.2.2 Hospital and Healthcare Service Search

Field	**Description**
Use Case Name	Hospital and Healthcare Service Search
Primary Actor	General User
Priority	High
Description	Allows users to search hospitals, departments, and diagnostic services.
Main Success Flow	1. User opens the search option. 2. User enters a hospital name, location, department, or service. 3. System searches the healthcare facility database. 4. System retrieves matching information. 5. System displays facility details. 6. User selects a facility. 7. System displays its location and map directions.

7.2.3 Medical Report Explanation

Field	**Description**
Use Case Name	Medical Report Explanation
Primary Actor	General User
Priority	High

Description	Allows users to upload medical reports and receive simple Bangla explanations.
Main Success Flow	1. User selects Medical Report Explanation. 2. User uploads a medical report image or PDF. 3. System validates the file. 4. OCR/Vision extracts readable text. 5. System sends the extracted text to the LLM. 6. LLM generates a simple Bangla explanation. 7. System displays the explanation with a medical safety notice.

7.2.4 Healthcare Facility Management

Field	**Description**
Use Case Name	Healthcare Facility Management
Primary Actor	System Administrator
Priority	High
Description	Allows administrators to add, update, and manage healthcare facility records.
Main Success Flow	1. Administrator opens the admin panel. 2. Administrator logs in using authorized credentials. 3. System verifies the credentials. 4. Administrator accesses facility management. 5. Administrator adds or updates facility records. 6. System validates and saves the information. 7. System displays a confirmation message.

7.3 Activity Diagram: Medical Report Explanation Process

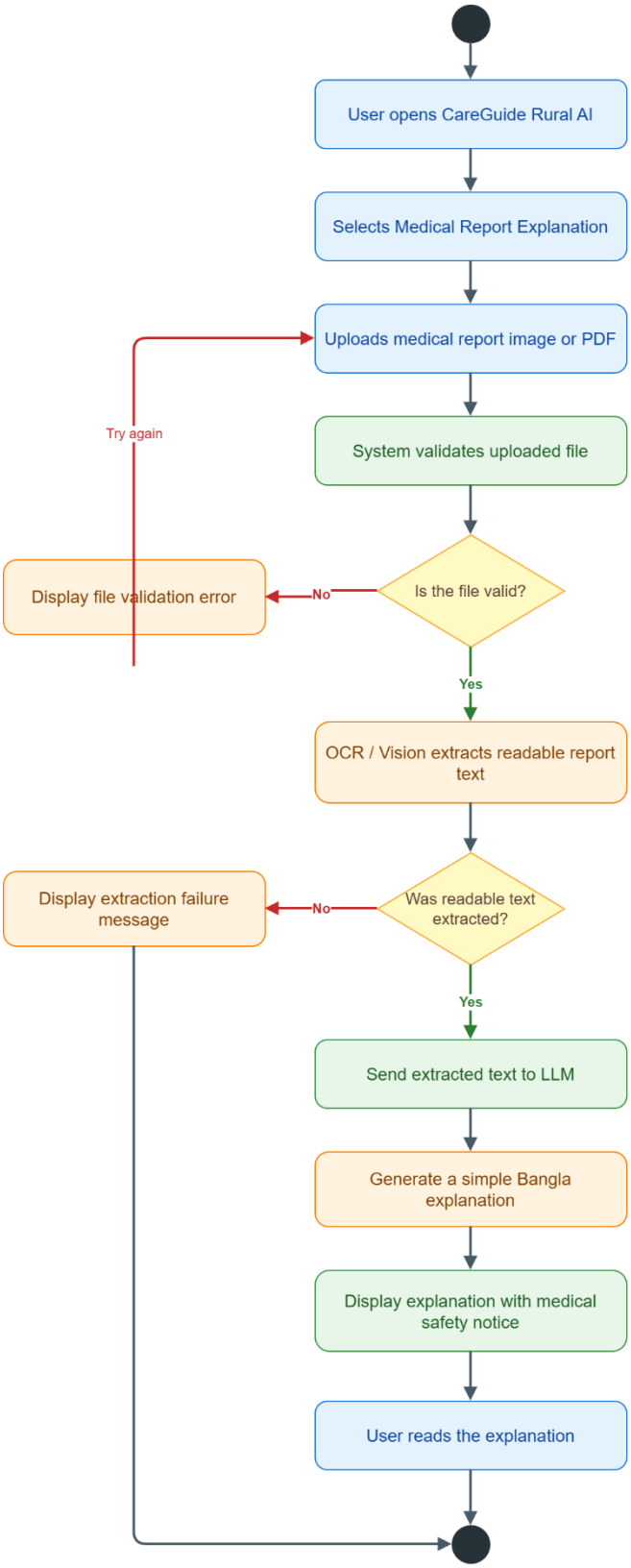
This Activity Diagram maps out the procedural flow of the Medical Report Explanation feature, showing how a user uploads a medical report, how the system validates the file, extracts readable text using OCR/Vision, and generates a simple Bangla explanation using an LLM. It also illustrates the decision-making process for invalid files and failed text extraction, including the corresponding error-handling steps.

Step-by-Step Flow:

1. User opens CareGuide Rural AI.
2. User selects Medical Report Explanation.
3. User uploads a medical report image or PDF.
4. System validates the uploaded file.
5. If the file is invalid, the system displays a validation error and asks the user to try again.
6. If the file is valid, OCR/Vision extracts readable report text.
7. If readable text is not extracted, the system displays an extraction failure message.
8. If text is extracted successfully, the system sends it to the LLM.
9. LLM generates a simple Bangla explanation.
10. System displays the explanation with a medical safety notice.
11. User reads the explanation.
12. The process ends.

7.3 Activity Diagram

Medical Report Explanation Process



7.4 Entity Relationship Diagram (ERD)

The ERD defines the database schema of CareGuide Rural AI by identifying entities, their attributes, primary keys (PK), foreign keys (FK), and relationships.

Main Entities:

Users, Message, Announcement, MedicalDocument, ReportExtraction, ExplanationHistory, SearchHistory, Notification, HealthcareFacility, Department, DiagnosticService, EmergencyInfo, Source, KnowledgeBase, ErrorReport, AdminReview, and UncertaintyNotice.

Key Relationships:

One-to-Many (1:N):

1. One User can have many Messages (1:N).
2. One User can upload many MedicalDocuments (1:N).
3. One MedicalDocument can generate multiple ReportExtractions (1:N).
4. One User can have many SearchHistory records (1:N).
5. One HealthcareFacility can have many Departments (1:N).
6. One HealthcareFacility can offer many DiagnosticServices (1:N).
7. One Source can provide many KnowledgeBase records (1:N).
8. One ErrorReport can have multiple AdminReviews (1:N).

Many-to-One (N:1):

1. Many Messages belong to one User (N:1).
2. Many MedicalDocuments belong to one User (N:1).
3. Many ReportExtractions belong to one MedicalDocument (N:1).
4. Many SearchHistory records belong to one User (N:1).
5. Many Departments belong to one HealthcareFacility (N:1).
6. Many DiagnosticServices belong to one HealthcareFacility (N:1).
7. Many KnowledgeBase records belong to one Source (N:1).
8. Many AdminReviews belong to one ErrorReport (N:1).

Many-to-Many (M:N):

- One User can view many Announcements, and one Announcement can be viewed by many Users (M:N).

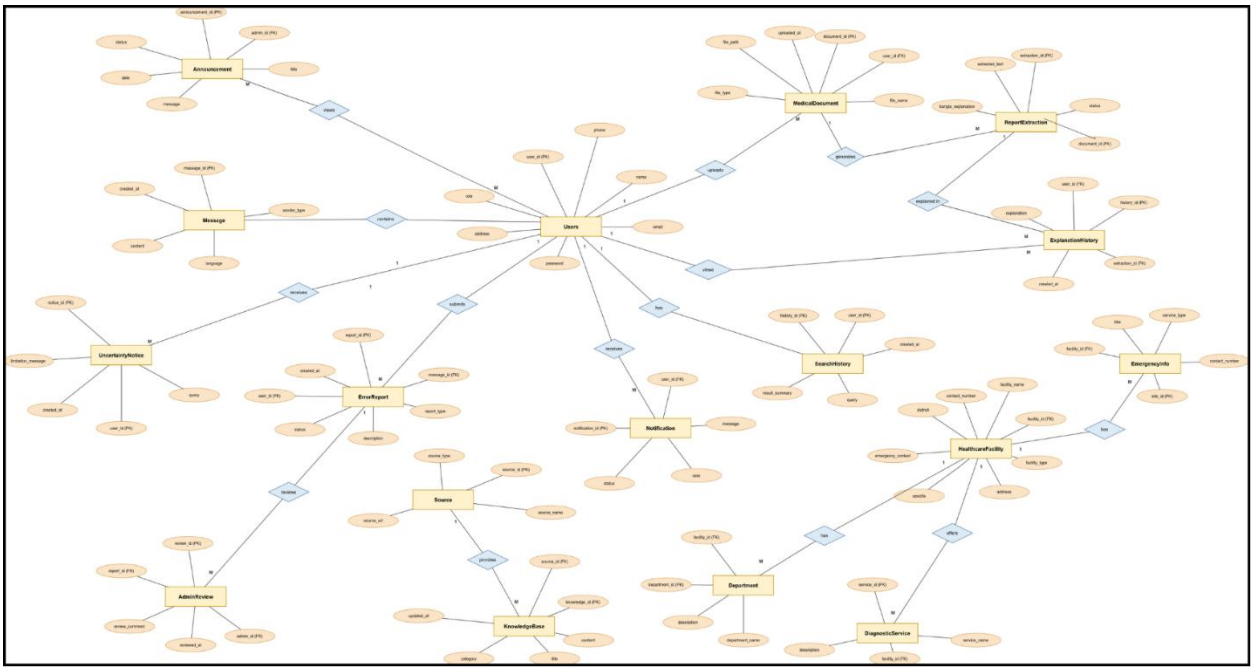


Fig: Entity Relationship Diagram (ERD)

7.5 AI Feature Design

CareGuide Rural AI will use Gemini 3.8 Flash as the primary candidate LLM and Gemini 3.5 Flash Lite as a fallback model. Gemini Function Calling and LangGraph will manage tool selection and the AI agent workflow. BGE-M3 embeddings, RAG, and PostgreSQL with pgvector are planned for healthcare information retrieval. Gemini 3.5 Transcribe and Gemini 3.8 Flash TTS have been tested for Bangla voice input and output. EasyOCR and Gemini are planned for medical report text extraction and explanation. These components will be integrated through a Python FastAPI backend.

The system will use pretrained models rather than training an LLM from scratch. It will provide general healthcare information and help users locate healthcare facilities, but will not diagnose diseases, prescribe medicines, or replace qualified healthcare professionals.

7.5.1 Data

The system will use two main types of data:

- **Healthcare Knowledge Data**
- **Healthcare Facility Data**

Users may also upload medical report images or documents for explanation.

A. Healthcare Knowledge Data

- **Source:** Trusted public health information from government health authorities and recognized healthcare organizations.
- **Content:** General health information, preventive guidance, common healthcare questions, and medical terminology explanations.
- **Size:** A curated collection of verified documents will be prepared according to the project's scope.
- **Metadata:** Each document will include available metadata such as title, publisher, source URL, topic, and review date.
- **Preparation:** Documents will be cleaned, converted into text, and divided into smaller chunks. BGE-M3 embeddings will be generated and stored in PostgreSQL using pgvector for semantic retrieval in the planned RAG pipeline.

B. Healthcare Facility Data

- **Source:** Reliable public sources and verified healthcare facility records.
- **Content:** Facility name, address, location, contact number, departments, diagnostic services, coordinates, and verification date, where available.
- **Size:** The initial dataset will cover a selected geographic area rather than all healthcare facilities in Bangladesh.
- **Metadata:** Records will be categorized by facility type, location, department, and available services. Missing or unverified information will be identified.
- **Preparation:** Duplicate records will be removed, addresses and service names standardized, and available information checked before storage in PostgreSQL. Administrators will update records when verified information becomes available.

C. User-Provided Medical Reports

Users may upload readable medical report images or documents for text extraction and explanation. These reports will not be used to train a new model. Uploaded files and extracted content will follow the application's privacy and retention rules. Explanations will be saved in **Report_History** only with user consent.

No fixed dataset size or labelled clinical dataset is assumed. The actual data volume will be documented after initial data collection and preparation.

7.5.2 Model and Approach

The system will use pretrained AI models, API-based services, and AI integration frameworks rather than developing models from scratch.

AI Component	Selected Model / Approach	Purpose
Language understanding and response generation	Gemini 3.8 Flash (primary candidate) and Gemini 3.5 Flash Lite (fallback)	Understand user queries and generate responses in simple Bangla
AI application integration	LangChain	Integrate the LLM, prompts, and retrieval components
Healthcare knowledge retrieval	BGE-M3 embeddings with PostgreSQL/pgvector	Retrieve semantically relevant healthcare information
Answer grounding	Retrieval-Augmented Generation (RAG)	Generate answers based on retrieved healthcare information and provide source references
Agent workflow and request routing	Gemini Function Calling and LangGraph	Select predefined tools and manage the agent workflow
Medical report processing	EasyOCR and Gemini API	Extract report text and explain it in simple Bangla
Facility and service lookup	PostgreSQL queries	Retrieve matching healthcare facilities and services
Voice input (STT)	Gemini 3.5 Transcribe	Convert spoken Bangla queries into text
Voice output (TTS)	Gemini 3.8 Flash TTS	Convert generated Bangla responses into audio

The LLM, speech services, function calling, and a basic Gemini–LangGraph workflow have undergone initial testing. The RAG pipeline, real database queries, and complete medical report processing workflow remain planned for implementation.

Why is this approach suitable?

- **Gemini API:** Generates natural-language responses without training an LLM from scratch.
- **LangChain:** Simplifies integration of the LLM, prompts, and retrieval components.
- **BGE-M3 with pgvector:** Supports semantic retrieval of relevant healthcare information, with Bangla and Banglish performance to be evaluated.

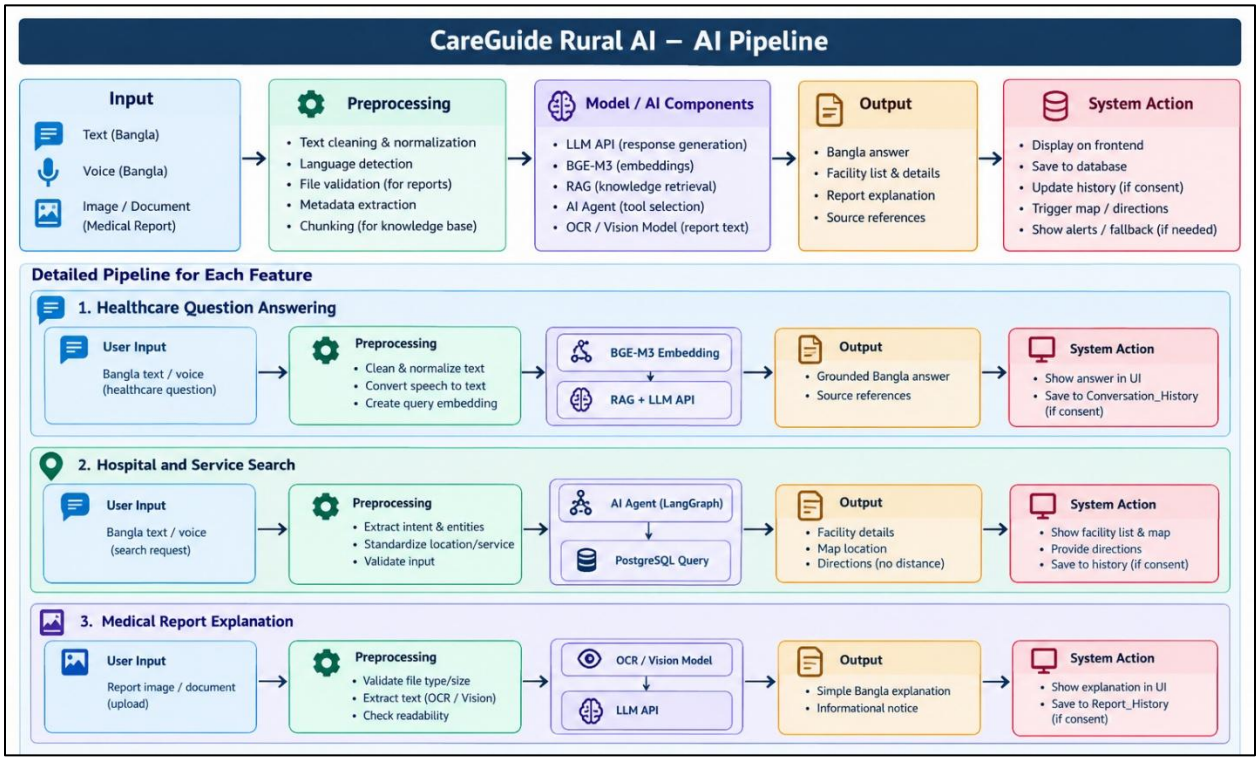
- **RAG:** Grounds responses in retrieved healthcare documents and supports source references.
- **LangGraph:** Manages the AI agent workflow, including tool selection and execution.
- **EasyOCR and Gemini API:** Support medical report text extraction and explanation in simple Bangla.
- **PostgreSQL:** Stores and retrieves structured hospital, department, and diagnostic service records.

Baseline Comparison

1. **Healthcare Question Answering:** Compare RAG-based responses with predefined FAQ or keyword-based retrieval using the same test questions.
2. **Facility Search:** Compare AI-assisted query interpretation with direct keyword-based database search.
3. **Medical Report Processing:** Compare OCR-extracted text with manually verified text and evaluate explanation quality separately.

These comparisons will be conducted during evaluation, and no method will be considered superior until testing is completed.

7.5.3 AI Pipeline



7.5.4 Integration with the Application

The React frontend will communicate with the FastAPI backend through REST APIs. The backend will coordinate AI services, database queries, and tool execution, returning results in JSON format.

The integration will support:

- **AI Question Answering:** Gemini API and the planned RAG pipeline will generate Bangla responses grounded in retrieved healthcare information.
- **Facility Search:** PostgreSQL queries will retrieve matching hospitals and healthcare services.
- **Report Explanation:** EasyOCR and Gemini API will extract and explain medical report information in simple Bangla.
- **Result Storage and Retrieval:** Authenticated users will be able to access their saved conversation history and report explanations, subject to user consent and access controls.
- **Security and Error Handling:** FastAPI will validate requests, handle AI service failures, protect API credentials, and restrict access to users' own saved data.

The complete backend integration and database operations remain to be implemented.

7.5.5 Evaluation and Fallback

Evaluation

- **Test Data:** Bangla healthcare queries, trusted healthcare documents, facility records, and verified medical report images.
- **Retrieval Performance:** Precision@k and Recall@k for evaluating RAG retrieval quality.
- **Answer Quality:** Manual review of factual accuracy, relevance, and source support.
- **OCR Performance:** Character Error Rate (CER) against verified report text.
- **System Performance:** Response time, success rate, and error rate.
- **Safety Testing:** Check for unsupported medical claims, diagnoses, prescriptions, and treatment recommendations.

Actual evaluation results will be reported after implementation and testing.

Fallback

- **Insufficient Information:** Inform users when reliable information is unavailable.
- **AI API Failure:** Use the tested fallback LLM where applicable; otherwise, display an error message.
- **Search Failure:** Inform users when no verified facility or service records are found.
- **OCR or Voice Failure:** Request a clearer image or allow text input where possible.
- **Database or Network Failure:** Display a user-friendly error without presenting unverified information as confirmed.

- **Unsafe Requests:** Explain system limitations and recommend consulting a qualified healthcare professional when appropriate.

These fallback mechanisms will be implemented and evaluated as development progresses.

8. Team Member Responsibility Matrix

The following table shows the primary responsibilities of each team member in the development of CareGuide Rural AI. All members also participate in testing, evaluation, and documentation.

Member Name	ID	Primary Responsibilities
Sharmin Akter Juthi	241-35-282	Responsible for backend API development using FastAPI, PostgreSQL/pgvector database design, authentication, Bangla voice processing, Speech-to-Text (STT), Text-to-Speech (TTS), hospital and facility search APIs, admin management, and knowledge base APIs.
Meherin Akter Mim	241-35-101	Responsible for frontend UI/UX development using React.js and Tailwind CSS, Bangla interface design, microphone interaction UI, audio playback interface, conversation and report history screens, and state management.
Tanjeem Rethnee	241-35-036	Responsible for AI component development, including LLM-based response generation, RAG pipeline implementation, BGE-M3 embeddings, Gemini and LangGraph Agent integration, Bangla voice processing, medical report text extraction using OCR, AI safety checks, and response evaluation.